

Load Cell Sediment Trap

September 28, 2026

Components

1. Outer aluminum box
2. Inner aluminum box
3. Box lid with cutout
4. Bolt and wingnuts to fix the lid to the outer box
5. Wooden box in the creek bed
6. Three waterproof load cells
7. Data logger

Important Design Considerations

1. The lid and inner box must be easily removed while submerged in water. This means no power tools necessary and ideally no tools at all.
2. The inner box has to be fixed to the load cells so that it does not lean on the outer box at all when sediment is added. All weight additions must be felt by the cells and only the cells. As far as I can tell, it is not possible to use less than three load cells without making a different box design.
3. Make room for wires in the outer wooden box.

Images with Annotations



